

LAB EXPERIMENTS

Q1) Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

Aim

To write a C program to implement the Caesar Cipher technique by replacing each letter in the plaintext with a letter that is k positions further down the alphabet (where k is between 1 and 25), and to generate the corresponding ciphertext.

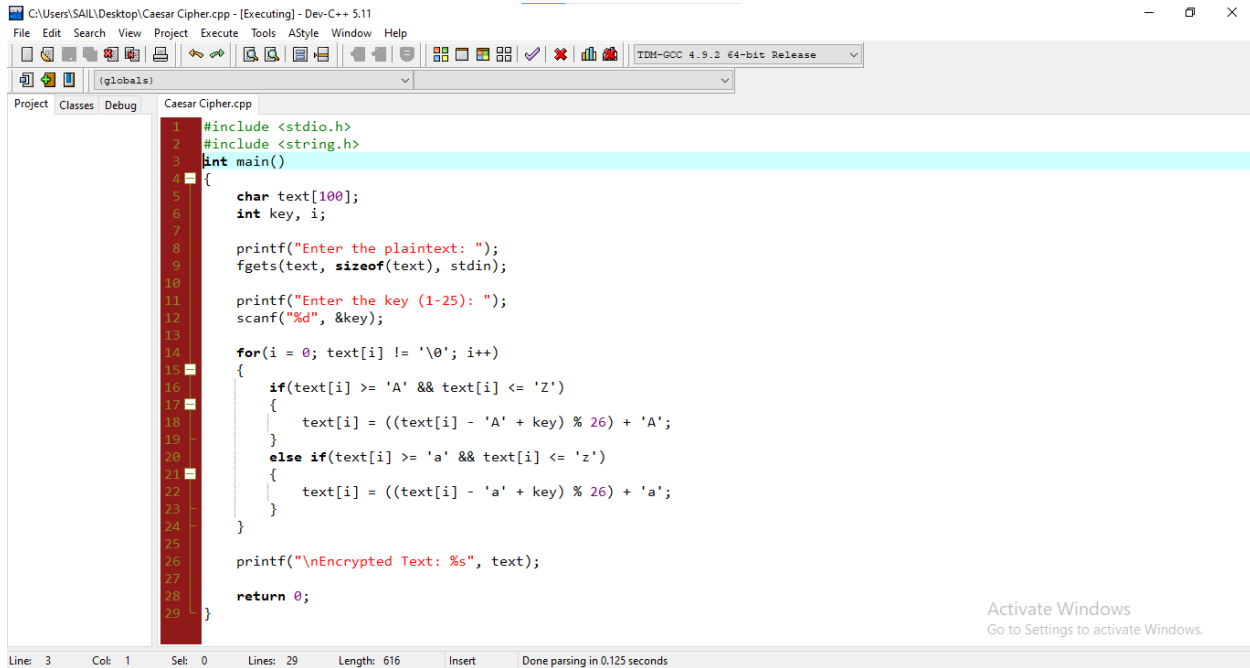
Algorithm

1. Start the program.
2. Declare variables for the plaintext, ciphertext, key (k), and loop counter.
3. Read the plaintext message from the user.
4. Read the encryption key (k) from the user (1–25).
5. Traverse each character of the plaintext.
6. If the character is an uppercase letter, shift it by k positions using modulo 26 and store it in the ciphertext.
7. If the character is a lowercase letter, shift it similarly while preserving lowercase format. Leave non-alphabetic characters unchanged.
8. Display the encrypted text (ciphertext).
9. Stop the program.

Result

Thus, the C program to implement the **Caesar Cipher** encryption technique was successfully executed, and the corresponding ciphertext was obtained.

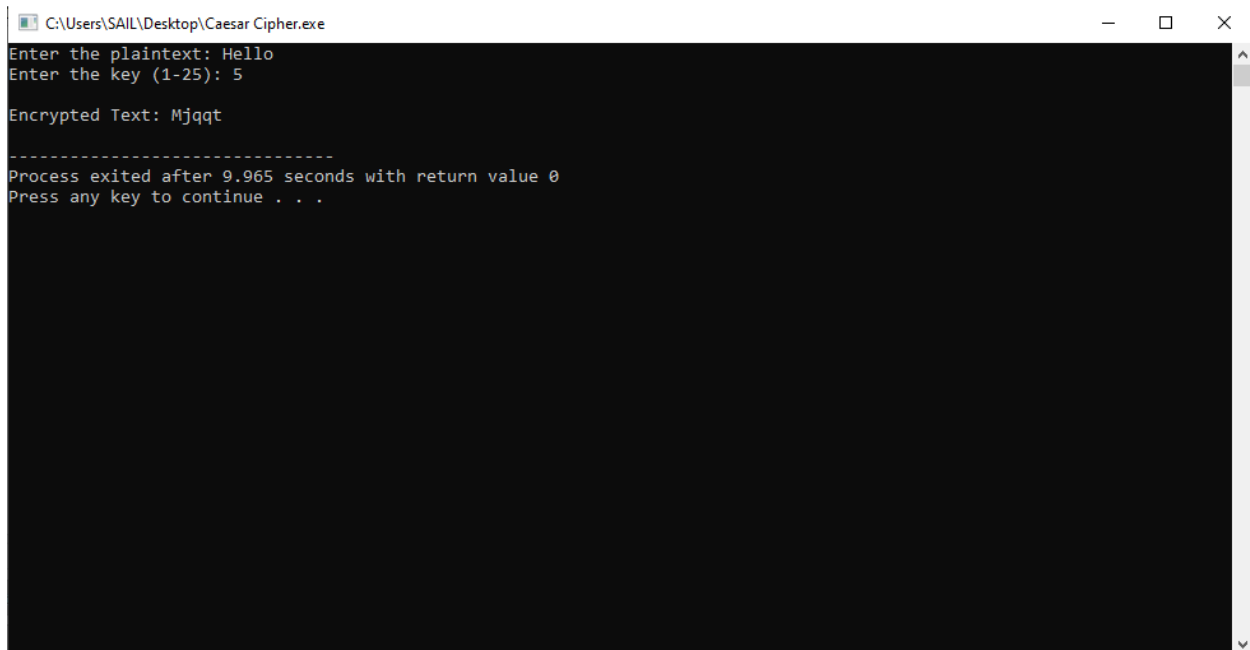
Code:



```
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char text[100];
6     int key, i;
7
8     printf("Enter the plaintext: ");
9     fgets(text, sizeof(text), stdin);
10
11     printf("Enter the key (1-25): ");
12     scanf("%d", &key);
13
14     for(i = 0; text[i] != '\0'; i++)
15     {
16         if(text[i] >= 'A' && text[i] <= 'Z')
17         {
18             text[i] = ((text[i] - 'A' + key) % 26) + 'A';
19         }
20         else if(text[i] >= 'a' && text[i] <= 'z')
21         {
22             text[i] = ((text[i] - 'a' + key) % 26) + 'a';
23         }
24     }
25
26     printf("\nEncrypted Text: %s", text);
27
28     return 0;
29 }
```

Line: 3 Col: 1 Sel: 0 Lines: 29 Length: 616 Insert Done parsing in 0.125 seconds

Output



```
C:\Users\SAIL\Desktop\Caesar Cipher.exe
Enter the plaintext: Hello
Enter the key (1-25): 5

Encrypted Text: Mjqqt

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Process exited after 9.965 seconds with return value 0
Press any key to continue . . .
```